Senior Applied AI Engineer — Product & Revenue Systems

Job description

Role Mission

Build and deploy AI systems that directly drive revenue by embedding AI across QEA's product, sales, and marketing workflows.

This role focuses on execution: turning QEA's building data, imagery, analysis outputs, and business workflows into production AI tools that accelerate deals, generate pipeline, improve proposal quality, and enable AI-assisted retrofit planning.

The ideal candidate is a senior hands-on builder who can work across product, engineering, sales, and business teams to identify high-impact AI opportunities, build working systems, and deploy them into real operational use.

What Success Looks Like: 12–18 Months

Product: Ship AI-assisted retrofit planning features, including detection, recommendations, prioritization, and ROI-supporting outputs.

Sales: Improve win rate and sales velocity through AI-generated insights, automated proposal support, portfolio-level analysis, and pre-sales intelligence.

Marketing / Growth: Develop AI-driven targeting, outbound, and campaign automation systems that generate qualified pipeline.

Platform: Help establish QEA as an AI-driven building intelligence platform that supports retrofit planning, financing, and long-term asset management.

Core Responsibilities

• Build AI-powered product features for retrofit planning, including building issue detection, retrofit recommendations, prioritization logic, and ROI-supporting outputs.

• Develop AI-assisted sales tools, including pre-sales insights, automated proposal generation, portfolio analysis, client-specific opportunity summaries, and supporting materials for business development.
• Create AI-driven marketing and growth systems, including lead targeting, account research, personalized outreach support, and campaign automation.
• Work with QEA's data pipelines, imagery, analysis outputs, and customer data to continuously improve AI systems and product capabilities.
• Support SaaS product development by contributing to APIs, dashboards, automation workflows, internal tools, and production AI integrations.
• Collaborate closely with product, engineering, building science, sales, and leadership teams to prioritize and deploy high-impact solutions quickly.
• Help define the technical roadmap for applied AI across QEA's product and revenue systems.
• Provide technical direction to junior or intermediate engineers as the AI team grows.

Ideal Candidate Profile

The ideal candidate has 5+ years of experience in applied AI, machine learning, data engineering, computer vision, LLM-based systems, or growth engineering.

They are a strong hands-on builder with experience shipping production systems, not just experiments or prototypes. They are comfortable working with ambiguous business problems, translating them into technical solutions, and iterating quickly based on real-world feedback.

This person should be capable of operating as a senior technical lead, helping QEA prioritize AI initiatives and eventually directing junior or intermediate team members as execution needs grow.

Required Technical Skills

Strong experience with:
• Python and modern AI / ML development workflows
• Machine learning frameworks and production model deployment
• Computer vision systems, especially image-based detection, segmentation, classification, or analysis
• LLM-based systems, including document processing, workflow automation, retrieval, structured extraction, or AI-assisted content generation
• Structured and unstructured data

• Data pipelines and model improvement workflows
• Deploying AI systems into production environments
• APIs, SaaS product development, dashboards, or internal automation tools

Nice-to-Have Experience

Experience with any of the following would be valuable:
• Building science, energy modelling, construction, retrofits, or real estate analytics
• Drone imagery, thermal imagery, geospatial data, or 3D building data
• Proposal automation, sales enablement tools, or portfolio analysis systems
• Marketing automation, lead scoring, or AI-driven outbound systems
• Startup or scale-up environments where speed, ownership, and ambiguity are part of the role

Mindset

Execution-focused builder.

This person should move quickly, prioritise impact, and be comfortable working across technical and business domains. They should be able to identify where AI can create real business value, build practical solutions, and avoid getting stuck in research mode when a usable system can be shipped and improved.

They should be comfortable with ambiguity, able to work directly with non-technical stakeholders, and capable of turning loose business needs into deployed products and revenue tools.

First 90-Day Plan

Month 1: Understand QEA's Data, Product, and Workflows
• Learn QEA's current product, data assets, AI workflows, sales process, and retrofit planning goals.
• Review existing imagery, analysis outputs, building data, proposals, and customer-facing deliverables.
• Identify the highest-impact AI opportunities across product, sales, and growth.
• Define a focused execution plan for the first phase of work.

Month 2: Build Initial High-Impact AI Tools
• Build and test initial AI-assisted sales or proposal tools.

• Begin developing core AI-assisted retrofit planning features.
• Work with engineering and product teams to integrate early tools into internal workflows.
• Validate outputs with sales, building science, and leadership teams.

Month 3: Launch First Production Workflows
• Launch the first usable AI-driven sales, proposal, or retrofit planning workflow.
• Measure impact on speed, quality, and business usefulness.
• Iterate based on feedback from internal users and customer-facing teams.
• Define the next phase of the roadmap, including whether additional junior or intermediate support is needed.

Role Scope and Team Structure

This role has a broad 12–18 month mandate across product AI, sales enablement, marketing automation, proposal generation, portfolio analysis, and retrofit planning.

Given the breadth of the opportunity, the role should initially focus on the highest-impact priorities rather than attempting to execute every workstream at once. The senior hire will help determine sequencing, technical approach, and resource needs.

As the work expands, this role may grow into a technical lead position with support from junior or intermediate engineers who can help execute under the senior engineer's direction.

Positioning

QEA is building the system that helps decide how buildings get fixed, financed, and managed.

This role sits at the center of that transition by turning QEA's building intelligence, AI capabilities, and customer workflows into scalable product and revenue systems.

Impact

This role directly impacts:
• Revenue growth
• Sales velocity
• Product differentiation

- Retrofit planning capabilities
- Proposal quality and automation
- AI-driven platform development
- QEA’s transition from services-enabled software toward a scalable AI-driven platform

The right person will help QEA move faster, sell smarter, and build the next generation of AI-powered building intelligence tools.